

College of Industrial Technology  
King Mongkut's University of Technology North Bangkok



Final Examination of semester 1

Year: 2013

Subject: 392133 Physics III

Section: 15-16

Date: 28 September 2013

Time: 08.00-10.00

Name: \_\_\_\_\_ ID: \_\_\_\_\_ Class: \_\_\_\_\_

Instructions

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken inside the examination room.
3. Textbook and dictionaries are **NOT** allowed.
4. A scientific calculator is allowed.
5. Express your answer in English. Other languages will be discarded.
6. Electronic communication devices are **NOT** allowed in the examination room.
7. The examination has 7 pages (including this page), 6 problems with the total scores of 60 points.
8. Write all your solution and answer on this examination sheet.

1.1 [2 points] The speed of light changes when it goes from ethyl alcohol ( $n = 1.36$ ) to carbon tetrachloride ( $n = 1.46$ ). Please determine the ratio of the speed in carbon tetrachloride to the speed in ethyl alcohol.

1.2 [2 points] Light (from air) is refracted through a diamond. If the angle of incidence is  $30^\circ$ , and the angle of refraction is  $12^\circ$ , what is the index of refraction?

1.3 [3 points] Aliens from a planet in another galaxy have their eyes at the exact center of their 4.0 m long bodies. How long must a plane mirror be for such a creature to be able to see all of its body in the mirror?

1.4 [3 points] The actual depth of a shallow pool 1.00 m deep is not the same as the apparent depth seen when you look straight down at the pool from above. How deep (in m) will it appear to be? ( $n_{\text{water}} = 1.33$ .)

2.1 A concave mirror has a focal length of 20 cm.

(a) [2 points] What is the position (in cm) of the object if the image is upright and is two times larger than the object?

(b) [1 point] Draw a ray diagram to confirm your calculation.

2.2 [3 points] An object 50-cm high is placed 1.0 m in front of a converging lens whose focal length is 1.5 m. Determine the image height (in cm).

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2.3 [4 points] An object is placed 25 cm in front of a lens of focal length 20 cm. However, 60 cm past the first lens is a second lens of focal length 25 cm. What is the resulting magnification of the object in this setup?

3.1 [3 points] A plan convex lens is made of glass ( $n = 1.5$ ) with one flat surface and the other having a radius of 20 cm. What are the focal length (in cm) and the power (in Diopter) of the lens?

3.2 [3 points] A far-sighted student has a near point of 1.0 m. Calculate the focal length (in cm) of the glasses needed so the near point will be normal (25 cm).

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3.3 [4 points] A 10-cm focal length converging lens is used to magnify small newspaper print 2.0 mm high. Calculate the height of the image (in mm) for a normal eye (25 cm).

4.1 [5 points] Two slits separated by 0.10 mm are illuminated with green light ( $\lambda = 540$  nm). Calculate the distance (in cm) from the central bright-region to the fifth bright band ( $m=5$ ) if the screen is 1.0 m away.

4.2 [5 points] A narrow slit is illuminated with sodium yellow light of wavelength ( $\lambda$ ) 589 nm. If the central maximum extends to  $\pm 30.0^\circ$ , how wide is the slit?

5.1 [5 points] A helium-neon laser ( $\lambda = 632.8 \text{ nm}$ ) is used to calibrate a diffraction grating. If the first-order maximum occurs at  $20.5^\circ$ , what is the number of lines per unit length (lines/cm),  $N$ ?

5.2 [5 points] Nonreflective coatings for camera lenses reduce the loss of light at various surfaces of multi-lens systems. Find the minimum thickness of a layer of magnesium fluoride ( $n = 1.38$ ) on flint glass ( $n = 1.66$ ) that will cause destructive interference of reflected light of wavelength  $\lambda = 550 \text{ nm}$ .



6.1 [5 points] The polarizing angle in air for diamond is  $67.4^\circ$ . Calculate the critical angle inside the diamond for total internal reflection.

6.2 [5 points] Three polarizing disks whose planes are parallel are centered on a common axis. The direction of the transmission axis in each case is shown in Figure relative to the common vertical direction. A plane-polarized beam of light with  $E_0$  parallel to the vertical reference direction is incident from the left on the first disk with intensity  $I_i = 10.0$  units (arbitrary). Calculate the transmitted intensity  $I_f$  when  $\theta_1 = 0$ ,  $\theta_2 = 30^\circ$ ,  $\theta_3 = 60^\circ$

